

## EDUCATION

### University of California, Berkeley

*B.S. in Mechanical Engineering and Electrical Engineering & Computer Science, GPA: 3.82*

*Berkeley, CA*

*Aug 2025 - Present*

- Relevant Coursework: Electronics for IoT, Mechatronics Design, Linear Algebra, Thermodynamics, Python, Java, CAD (SolidWorks)

## RESEARCH

### Multiphase Thermofluidics & Surface Nanoengineering Lab

*Undergraduate Researcher*

*Berkeley, CA*

*Aug 2025 - Present*

- Developed embedded sensor integration using Beckhoff PLC (TwinCAT) for real-time acquisition and closed-loop control of flow meters, pressure sensors, and thermocouples
- Implemented electro-thermal-mechanical control logic using piezoelectric actuator feedback loops for autonomous surface cleaning mechanisms
- Calibrated and validated high-speed imaging systems (FLIR thermal, Photron Fastcam) through software-hardware interface tuning for experimental precision
- Selected materials based on spectral and thermal performance targets, validating outcomes through controlled experimental protocols
- Operated FabLight Laser Cutter to manufacture etched copper plates for heat distribution.
- Used SolidWorks & OnShape to design motor housing for gantry systems.

### CUNY STEM Research Academy

*Student Researcher*

*Queens, NY*

*June 2022 - Aug 2022 & June 2023 - Aug 2023*

- Integrated Raspberry Pi 4 with GPS/IMU modules, environmental sensors, and onboard imaging for CubeSAT
- Wrote data-logging software in Python to interface with I2C peripherals and log sensor data in real time
- Manufactured enclosures with Creality Ender-3 Printer utilizing FDM techniques.

## ACTIVITIES

### CalSol | UC Berkeley Solar Vehicle Team

*Aeroshell Team Engineer*

*Berkeley, CA*

*Aug 2025 - Mar 2026*

- Designed and validated a detachable canopy latching mechanism using FEA to confirm rear pin load capacity and structural integrity under operational constraints
- Manufactured custom latch hardware interfacing with carbon fiber composite structures; collaborated across composites and integration teams for system-level compatibility

## PROJECTS

### IMU-Based Driver Monitoring System @ StarkHacks

[Learn More](#)

*West Lafayette, IN*

*Apr 2026 - Apr 2026*

- Rapidly prototyped and debugged hardware/software during 36-hour hackathon (Purdue University)
- Designed and built a dual-device embedded system using ESP32-S3 microcontrollers and MPU-6050 IMUs to detect driving behaviors (hard braking, acceleration, lane changes) without cameras
- Utilized FDM printing to create wearable earpieces and housing for electronics.
- Built remote audio feedback system for dynamics alerts through WebSockets, Web Audio API, and ElevenLabs

### Mathematical Competition in Modeling (MCM) 2025

[Paper](#)

*Queens, NY*

*2025 | International Competition*

- Built a multi-model predictive framework in Python for time-series forecasting using SVD-based anomaly detection and censored data techniques
- Received MAA Outstanding Winner Award
- Delivered a full technical report under a 5-day deadline in a 3-person team

## PROFESSIONAL SKILLS

- Soldering, CNC Machining, Laser Cutting, Python, Java, Javascript, SQL, Flask, SolidWorks, OnShape, 3D printing (FDM), Beckhoff PLC